

Erdős #1016: keep the witness components

Eighth structural draft, September 25, 2026

Status. A new proof candidate, not Lean-verified. We specialize the forest lemma to the witnesses actually used: unions of few cycles. Preserving those components removes the expanding-region estimate and the entire signed cut-tree descent. Deleted short cycles need no connecting paths. The final sharp asymptotic is unchanged; all finite estimates used below are proved here.

Write $\beta(G) = |E| - |V| + \text{comp}(G)$ and let $\mathcal{Z}(G)$ be its binary cycle space. Every feasible prescribed-boundary class has $2^{\beta(G)}$ members. For a uniform $\xi \in \mathcal{Z}(G)$ put

$$P_G(M) = \mathbb{P}(\xi \cap M \text{ is a forest}).$$

Regions are induced; $P_G(H) = P_G(E(H))$. Auxiliary graphs may have loops and parallel edges, with loops counted twice in degrees. All logs are binary except $\ln$; $\log^{(j)} t$ means j successive binary logarithms, and $\log_*$ stops at one. Let $h(n)$ be the minimum of $|E(G)| - n$ over pancyclic simple n -vertex graphs. We prove $h(n) = \log n + \log_* n + O(1)$.

For a nonnegative random variable Z with $\mu = \mathbb{E}Z > 0$, Cauchy–Schwarz gives

$$\mathbb{E}Z^2 \leq 2\mu^2 + K\mu \implies \mathbb{P}(Z = 0) \leq 1 - \frac{\mu^2}{\mathbb{E}Z^2} \leq \frac{1}{2} + \frac{K}{4\mu + 2K}. \quad (1)$$

1 Cut correlations

Lemma 1 (Exceptional cuts). *In a connected graph, let A be a connected region with cut size d . Among connected regions B disjoint from A whose zero-cut events satisfy*

$$\mathbb{P}(\delta A = \delta B = 0) > 2\mathbb{P}(\delta A = 0)\mathbb{P}(\delta B = 0),$$

no vertex-disjoint subfamily has more than d members.

Proof. Contract A and a disjoint subfamily of the B 's to vertices. The cycle-space projection is onto, so it preserves their joint cut law. For two vertices v, w , count as links the vw edges and components of $G - \{v, w\}$ attached to both. If there are ℓ links, rank subtraction gives the zero-cut correlation ratio $2^{\ell-1}$.

Fix v . Append its d incidences as terminal leaves to a spanning forest of $G - v$, and prune nonterminal leaves. Each link to w gives a terminal-containing branch at w . Thus a vertex with at least three links is a branch vertex of this pruned forest. There are at most d branch vertices. □

Corollary 2 (Small cuts). *For k vertex-disjoint connected cyclic regions with cuts at most d in a connected graph,*

$$P_G\left(\bigcup_i E(H_i)\right) \leq \frac{1}{2} + \frac{(d+1)2^d}{2k}. \quad (2)$$

Proof. Let E_i say that the i th cut is zero and its inner even set is nonempty. Its probability p_i is at least 2^{-d-1} . Given zero cuts, the inner even sets are independent; hence the pairwise correlation ratio is exactly the zero-cut ratio. By Lemma 1, each event has at most d exceptional partners in this disjoint family. For the other pairs use $\mathbb{P}(E_i E_j) \leq 2p_i p_j$, and for exceptional partners use $\mathbb{P}(E_i E_j) \leq p_i$. Thus $Z = \sum_i \mathbf{1}_{E_i}$ has

$$\mu \geq k2^{-d-1}, \quad \mathbb{E}Z^2 \leq 2\mu^2 + (d+1)\mu.$$

Each E_i precludes a forest, so (1) proves the claim. □

2 The forest estimate, directly at the witness

Lemma 3 (Few-component witness). *For all sufficiently large r , let G be a connected simple graph of rank r and let W be a nonempty union of cycles. Write $m = |E(W)|$ and count $c = \text{comp}(W)$ on its edge support. If*

$$m \leq \sqrt{r}, \quad c \leq \frac{\log r}{100},$$

then

$$P_G(E(G) \setminus E(W)) \leq \frac{1}{2} + \frac{20}{\log^{(3)} r}. \quad (3)$$

Proof. Put $x = \log r$, $y = \log x$, $z = \log y$, $D = \lfloor x/10 \rfloor$ and $L = \lfloor x\sqrt{y} \rfloor$. Suppose the probability exceeds $1/2 + \varepsilon$, where $\varepsilon = 20/z$.

Cubicize without disconnecting the witness. Replace a vertex of degree $d > 3$ by a path of $d - 2$ cubic vertices, with two old incidences at each end and one at every internal vertex. If $k \geq 2$ incidences belong to W , put them first along the path and mark the initial segment meeting them. This segment is connected and has at most k vertices. Mark unchanged witness vertices too. In the expanded graph G' , the witness lifts to a subgraph on a marked set P with

$$|P| \leq 2m, \quad \text{comp}(G'[P]) \leq c.$$

Contracting the splitting trees is a cycle-space bijection. An original forest, lifted and augmented by all tree edges, is still a forest; hence restricting to the complement of P can only increase its probability. Prune unmarked leaves and suppress unmarked degree-two vertices. Their even coordinates are respectively zero or equal along a path, so the cycle-space law and the forest event off P are preserved. No marked vertex is removed; suppression may join marked components but cannot split them.

Call the resulting connected multigraph Γ . It has rank r , maximum degree three, every unmarked vertex is cubic, and

$$|P| \leq 2\sqrt{r}, \quad \text{comp}(\Gamma[P]) \leq c, \quad |\Gamma| = 2r + O(\sqrt{r}), \quad P_\Gamma(\Gamma - P) > \frac{1}{2} + \varepsilon. \quad (4)$$

The order identity follows from $|\Gamma| = 2r - 2 + \sum_v (3 - \deg_\Gamma v)$; only marked vertices contribute to the sum.

Delete short cycles, without connecting them. If $\Gamma - P$ contains $\lceil \sqrt{r} \rceil$ disjoint cycles of length at most D , including loops and digons, their induced regions have cuts at most D . Equation (2) gives $P_\Gamma(\Gamma - P) \leq 1/2 + O(xr^{-2/5})$, a contradiction. Otherwise choose a maximal such packing and let S be the union of its vertex sets. Then $|S| \leq D\sqrt{r}$, and

$$J = \Gamma - (P \cup S)$$

is simple with girth greater than D , order $v = (2 + o(1))r$, and deficit

$$d_J := \sum_{u \in J} (3 - \deg_J u) \leq 3(|P| + |S|) = O(\sqrt{r} \log r). \quad (5)$$

Neither P nor $P \cup S$ is required to be connected.

Supply cycle weight. For the directed-edge nonbacktracking matrix B , adjacency matrix A_0 and degree matrix Δ of J , put $M(t) = t^2 I - tA_0 + \Delta - I$. Then

$$M(2) = 2(\Delta - A_0) + 3I - \Delta \succeq 0, \quad \mathbf{1}^T M(2 - d_J/v) \mathbf{1} = 0.$$

Continuity gives a singular $M(t)$ with $t \geq 2 - d_J/v > 1$. A null vector u gives the nonzero B -eigenvector $f(i, j) = tu_j - u_i$. Conversely, a nonreal eigenvalue of B gives $M(t)u = 0$ with

$u \neq 0$; multiplication by u^* gives a real quadratic with constant-to-leading ratio at most two, so $|t| \leq \sqrt{2}$. Even powers of real eigenvalues are nonnegative. Therefore

$$2^{-\ell} \operatorname{tr} B^\ell \geq (1 - d_J/(2v))^\ell - 3v2^{-\ell/2} \quad (\ell \text{ even}).$$

By (5), summing over even $6x \leq \ell \leq L$ gives

$$T := \sum_{\ell \leq L} \frac{\operatorname{tr} B^\ell}{2^\ell 2^\ell} \geq (1/4 + o(1)) \ln \frac{L}{6x} = (1/8 + o(1)) \ln y.$$

Put $s = \lfloor (D-1)/2 \rfloor$. Girth makes an s -step nonbacktracking path between fixed endpoints unique. Thus the ordinary nonbacktracking walk count satisfies

$$2^{-t} N_t(u, w) \leq \frac{3}{2} 2^{-s} \quad (t > s). \quad (6)$$

Let $V = \sum_{C \subset J, |C| \leq L} 2^{-|C|}$, over simple cycles. A nonsimple cyclically nonbacktracking walk contains a proper simple cycle; its complementary ordinary closed nonbacktracking walk has length greater than D . Encode by the cycle, an oriented start, the complement and the original root. The factor $1/(2^\ell)$ cancels orientations and root positions; the start vertex and complementary length contribute at most L^2 . Consequently

$$0 \leq T - V \leq \frac{3}{2} L^2 2^{-s} V, \quad V \geq (1/8 + o(1)) \ln y. \quad (7)$$

Normalize cycle indicators. For each cycle $C \subset J$ of length at most L , let I_C say that C is a component of the random even set. Put

$$a_C = 2^{-|C|}, \quad p_C = \mathbb{P}(I_C), \quad X_C = (a_C/p_C) \mathbf{1}_{I_C}, \quad Z = \sum_C X_C.$$

Cubic ambient degree makes I_C exactly the event that all cycle edges are selected. These are feasible binary linear conditions, so $p_C \geq a_C$, $0 \leq X_C \leq 1$ and $\mathbb{E} X_C = a_C$.

For disjoint cycles C, C' , let $\ell(C, C')$ count their direct joining edges and the components of $\Gamma - (V(C) \cup V(C'))$ attached to both. Prescribing the inner cycle words multiplies zero-cut probabilities by separate inner factors. They cancel in the ratio from Lemma 1, giving

$$\mathbb{E}(X_C X_{C'}) = a_C a_{C'} 2^{\ell(C, C')-1}. \quad (8)$$

The deleted cycles count as cyclic components. At most c components of $\Gamma - (V(C) \cup V(C'))$ meet P , by (4). Every other component F_i lies in $\Gamma - P$ and has cut d_i ending on the two cycles, with $\sum_i d_i \leq 2L$. Set $t_0 = \lfloor y/2 \rfloor$. At most $2L/t_0$ components have $d_i > t_0$. Among the remaining components, at most $O(t_0 2^{t_0}/\varepsilon)$ are cyclic, by (2) and (4).

If F_i is a tree, it cannot meet S : any packed cycle meeting F_i is wholly contained in it, since it is disjoint from C, C' . Thus every such tree lies in the high-girth graph J . Cubic ambient degree gives $|F_i| = d_i - 2$. When it meets both cycles and $d_i \leq t_0$, choose a joining path through it of length at most t_0 ; include direct joining edges as paths of length one. These paths have disjoint interiors and distinct endpoints. Partition each cycle into consecutive vertex blocks of size at most $\lfloor D/4 \rfloor$. Two paths joining the same block pair would form a cycle of length at most $2t_0 + D/2 < D$. Hence there are only $O((L/D)^2)$ such paths. Uniformly in the pair,

$$\ell(C, C') \leq c + O\left((L/D)^2 + L/t_0 + t_0 2^{t_0}/\varepsilon\right) = c + o(x). \quad (9)$$

Finish the same second moment. Call a disjoint pair exceptional when $\ell(C, C') \geq 3$. By Lemma 1, a disjoint packing of exceptional partners of C has at most $|\delta C| \leq L$ members. A

maximal packing has union U_C of size at most L^2 , meeting every exceptional partner. Equation (6) bounds cycle weight through a fixed vertex by $O(L2^{-s})$. Thus (8)–(9) give

$$\sum_{C' \text{ exceptional to } C} \mathbb{E}(X_C X_{C'}) \leq a_C 2^{c+o(x)} O(L^3 2^{-s}) = o(1) a_C, \quad (10)$$

since $c \leq x/100$ and $s = x/20 + O(1)$. Intersecting distinct cycles are incompatible, nonexceptional disjoint pairs contribute at most $2a_C a_{C'}$, and $\mathbb{E}X_C^2 \leq a_C$. Therefore, with $\mu = \mathbb{E}Z = V$,

$$\mathbb{E}Z^2 \leq 2\mu^2 + (1 + o(1))\mu.$$

Equations (1) and (7) yield

$$P_\Gamma(\Gamma - P) \leq \mathbb{P}(Z = 0) \leq \frac{1}{2} + \frac{2 + o(1)}{\ln y} < \frac{1}{2} + \frac{20}{z},$$

contradicting (4). All estimates are uniform under the two witness bounds. $\square$

3 Cycle lengths and the conclusion

Let $\Phi(R)$ be the supremum of $(L - 2)2^{-\beta(G)}$ over simple graphs of rank at least R containing every cycle length $3, \dots, L$. Cycle counting gives $\Phi \leq 1$; vertex identifications make realizers connected. Fix a large integer R , and consider such a connected graph of rank $r \geq 2^{256R}$ with $L \geq 2^{2R} + 1$. Greedily adjoin a least-missing-length cycle until its union W has rank $t \geq 2R$. At preceding rank $s < 2R$, the added length is at most $2^s + 2$, and at most $\phi 2^s + 3$ when $s \geq R$, where $\phi = \Phi(R)$. Each step increases rank, so at most $2R$ cycles are used and $\text{comp}(W) \leq 2R$. With $m = |E(W)|$,

$$m \leq 2^{2R} + 4R - 1 \leq 2^{3R}, \quad m + 1 \leq \phi 2^{2R} + 2^R + 5R, \quad (m + 1)2^{-t} \leq \phi + 2^{1-R}.$$

Restriction to $M = E \setminus W$ has 2^{r-t} equally likely words. A mixed cycle requires a forest outside word and has at most $m + 1$ lengths per word. Pure outside cycles contribute at most one length per word, and inside cycles at most m lengths. Hence

$$(L - 2)2^{-r} \leq (m + 1)2^{-t} P_G(M) + 2^{-t} + m2^{-r}.$$

Lemma 3 applies: $m \leq \sqrt{r}$ and $\text{comp}(W) \leq 2R \leq (\log r)/100$. Since $\log^{(3)} r \geq \log^{(2)} R$, it proves

$$\Phi(2^{256R}) \leq \left(\frac{1}{2} + \frac{20}{\log^{(2)} R} \right) \Phi(R) + 2^{2-R}. \quad (11)$$

The omitted case $L < 2^{2R} + 1$ is below the error.

Iterate $R_{j+1} = 2^{256R_j}$. For $j \geq 2$, $\log^{(2)} R_j = 8 + 256R_{j-2}$, so both $\sum_j 1/\log^{(2)} R_j$ and $\sum_j 2^{j-R_j}$ converge. Equation (11) yields $\Phi(R_j) = O(2^{-j})$. Also $\log_* R_j = j + O(1)$: compare with $U_{j+1} = 2^{U_j}$ and induct on $R_j \leq \sqrt{U_j}$, starting with large fixed U_0 ; the reverse bound follows from $R_{j+1} \geq 2^{R_j}$. Monotonicity gives $\Phi(r) = O(2^{-\log_* r})$. A pancyclic n -vertex graph with $n + h$ edges has $r = h + 1$; hence $n - 2 = O(2^{r-\log_* r})$ and $n \leq 2^{r+1}$ imply $h \geq \log n + \log_* n - O(1)$.

Matching construction. On an n -cycle, $n \geq 7$, choose maximal k with $2^k + k + 1 \leq n$. An arc with successive segment lengths $2^i + 1$, $0 \leq i < k$, has endpoints $v_0, \dots, v_k$. Add the k segment shortcuts and $v_0 v_k$. Binary subset sums give $[k + 1, 2^k + k]$ and $[n - 2^k + 1, n]$, leaving at most one intermediate length, supplied by one chord. Starting at $t = k$, while $t > 2$ add $v_0 v_j$ for $j = \lceil \log t \rceil < t$. Its shortcuts give $[j + 1, 2^j + j]$; replace t by j . This takes $\log_* k - 1$ further chords, at most $\log n + \log_* n + O(1)$ altogether. Small n is harmless. Therefore

$$h(n) = \log_2 n + \log_* n + O(1).$$